

Semantic Drift



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Overview of the task

Language changes over time, especially in online communities where new slang and meanings emerge rapidly. This project investigates **semantic drift**, which refers to changes in how words are used or understood across time.

The system analyzes both **historical text corpora and modern social media data** to detect how word meanings evolve.

Research Questions

Can combining **historical text corpora (news articles)** with **modern social media data (Reddit)** reveal meaningful linguistic evolution in word usage?

Do modern internet communities **replace older meanings of words or introduce additional meanings?**

How can we automatically detect **semantic drift** using contextual embeddings?

Motivation

Language evolves continuously, especially in digital communities where new expressions, slang, and meanings spread quickly.

For example :

Word	Historical Meaning	Modern Meaning
Cloud	Weather phenomenon	Cloud computing
Tablet	Medicine	Digital Device

Traditional linguistic research studies semantic change manually using historical texts.

However, online communities generate new language extremely fast!

Detecting when meaningful linguistic changes occur can help determine when ML models need retraining.

Contributions

Data Preparation

Responsible for preparing and organizing the datasets used in the project.

- Collected and prepared Reddit datasets from target communities for new meaning sentences.
- Implemented the data preprocessing pipeline for Reddit comments.
- Used subreddits :
 - AskReddit
 - Youtube
 - GenZ
 - GenAlpha
 - Decadeology
- Used news dataset for old meaning sentences from Kaggle.

Preprocessing:

- Cleaned text data by removing URLs, special characters, and normalizing text.
- Structured the Reddit data into monthly JSONL files to support temporal analysis.
- For news dataset, downloaded and extracted word frequency timelines.
- Designed the project pipeline

Semantic Analysis

Responsible for implementing the core analysis pipeline and semantic drift detection system.

Main contributions:

- Implemented Sentence Transformer embeddings to represent contextual meaning.
- Implemented the old vs new meaning classification using K-Means Clustering.
- Developed visualization system using UMAP.

Evaluation:

Cosine similarity -> Drift score

Silhouette Score and Noise Ratio

Literature Review

Several studies have explored semantic change in language using contextual embeddings.

- **Hamilton et al. (2016)** studied diachronic word embeddings to measure semantic change over time using historical corpora. Their work introduced quantitative methods for measuring semantic drift.
- **Kutuzov et al. (2018)** provided a comprehensive survey of computational approaches for detecting lexical semantic change using neural embeddings and temporal corpora.
- **Garg et al. (2018)** demonstrated that contextual word embeddings can capture temporal linguistic shifts and societal biases across decades of text.

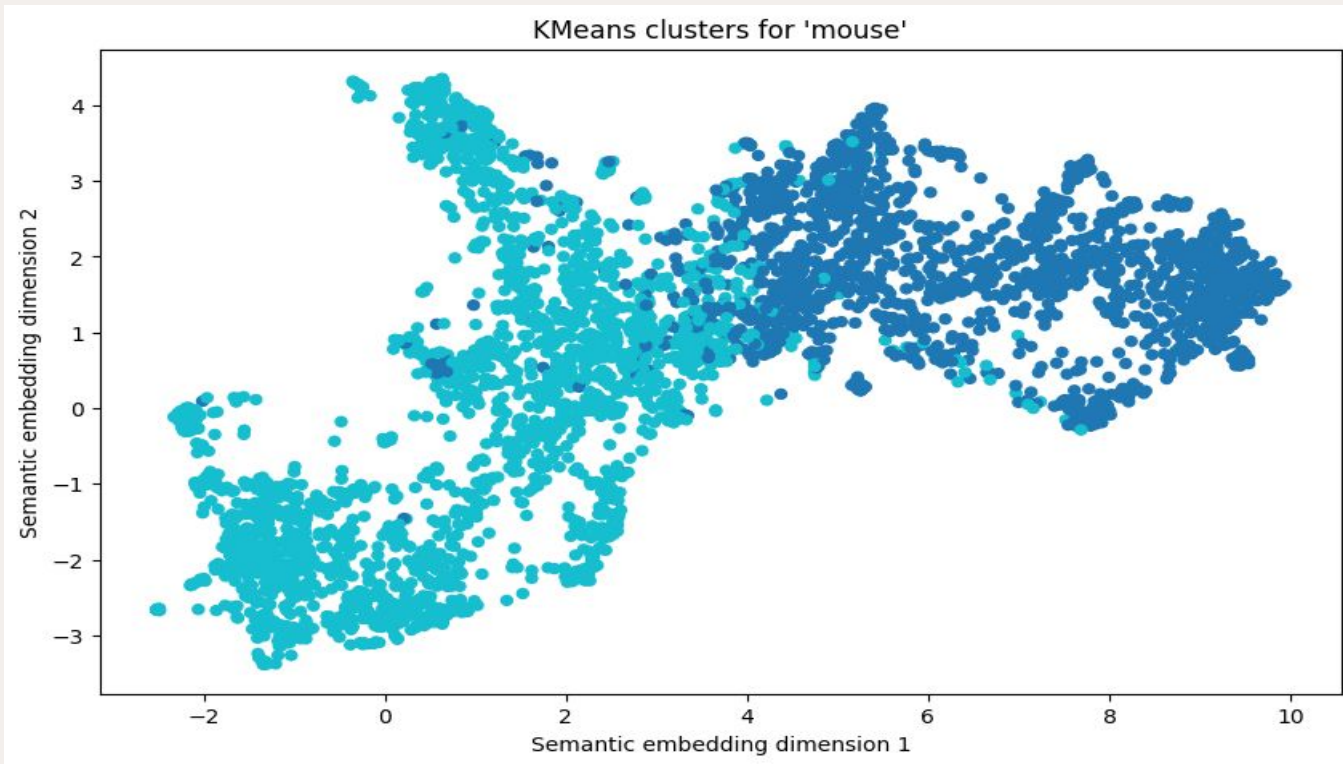
Project Pipeline



Why K-Means :

K-Means cleanly separates sentence embeddings into a small number of interpretable clusters, where each cluster corresponds to a different meaning of the word.

Results and evaluation metrics



Number of contexts analyzed: *mouse*: 1500 (news) + 2663 (Reddit)

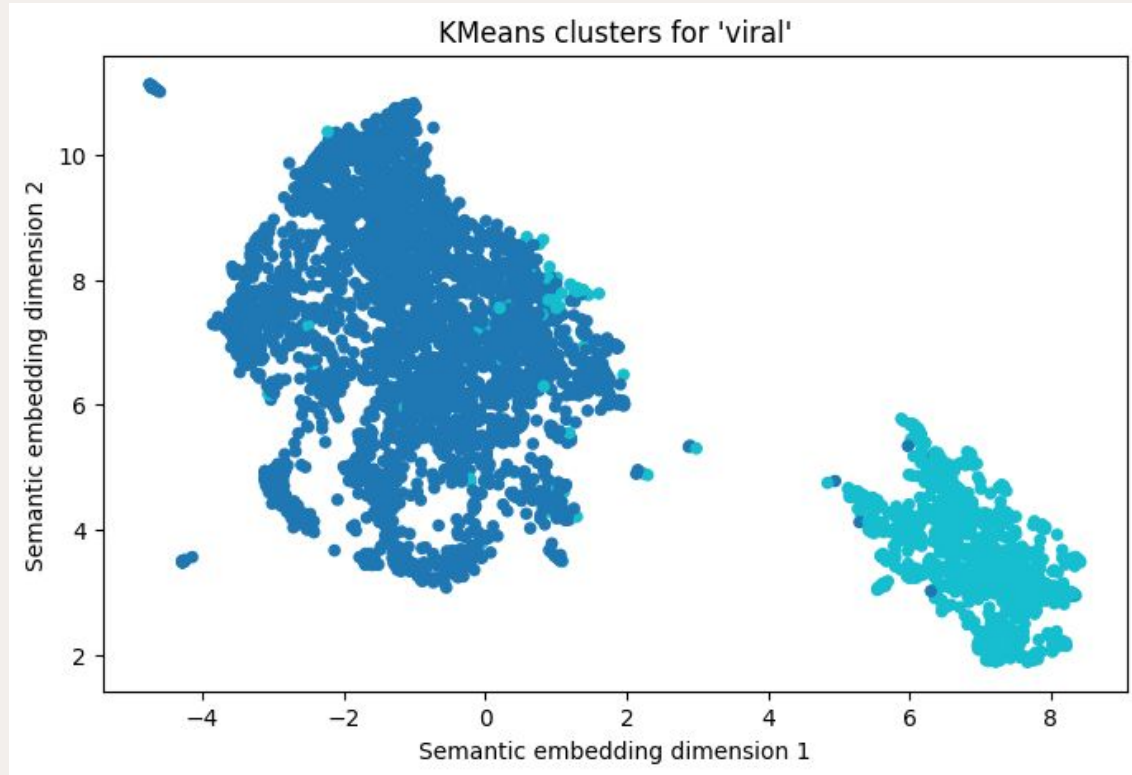
Drift Score : *mouse*: **0.125** (partial/moderate drift), which means both meanings exist.

Cluster 0

- But end-users have traditionally downloaded megabytes of information from the web, while uploading only kilobytes of key strokes and mouse clicks.
- Finally, when Google shows you its suggested driving route on the map, you can drag that line onto another road with your mouse.
- We then watch a video of a driver's-eye view of a road (somewhere in Reading, according to the signposts) and click our mouse when we see a hazard.
- Internet Explorer rather redundantly displays the "alt" tag when the mouse is hovered over the image; Firefox doesn't bother because no alternative is needed and there is a 'title' tag for captions.
- Now businesses are increasingly using networking tools such as LinkedIn and Facebook to identify those people with the click of a mouse.
- The Air Mouse is a rechargeable cordless mouse that can control a windows PC by being moved through the air or on a flat surface.
- They're tracking where your mouse is on the page, what you put in your shopping cart, what you don't buy.
- Review: Logitech VX Nano Cordless Laser Mouse praise-worthy
- If you miss the scroll bar found on almost every Windows laptops, the two-finger scroll option works well (run two fingers down the touch pad, and it scrolls like a mouse wheel).

Cluster 1

- An Oklahoma State University lab in Stillwater in December could not account for a dead mouse inoculated with bacteria that causes joint pain, weakness, lymph node swelling and pneumonia.
- The only time I thought "what's she got me into now?" was when I was ice skating, holding hands with Minnie Mouse.
- Across the park, hot dog vendors sought refuge under the palm trees" slowly shifting shadows, their carts painted with slap-dash cartoon characters like a Mickeyesque mouse and a clumsily drawn Pokémon.
- Obama Gobbles Down Mouse _ Made of Candy
- Importantly, in the presence of HIV, ZFN-modified CD4 T-cells also preferentially expanded in a mouse model.
- Mouse virus link to breast cancer
- After 40 minutes or so, the upstairs contingent was escorted to a smaller room below, where Mr. Roccoli, dressed in white pants and a grubby Mickey Mouse T-shirt (unbecoming indeed), performed a solo to more extracts from interviews and some uncredited songs.
- In one neighborhood of Rawalpindi, police played a cat-and-mouse game with the protesters.
- Christmas cracker contains dead mouse
- She may be 79 years old, but Minnie Mouse inspires very up to the minute designers.



Number of contexts analyzed: *viral*: 1500 (news) + 2787 (Reddit)

Drift Score : *viral*: **0.289** (high drift), which means new meaning dominates.

Cluster 0

- Doing the 'Aunt Jackie': A Harlem-filmed rap video goes viral on MySpace.
- Our launch program is built on a Viral Marketing and college grassroots strategy to capitalize on the inherent appeal that we believe ThePeeq.com will have with their audience.
- Viral events are unpredictable.
- By some estimates, the Star Wars kid is the most popular viral video ever, and viewed over one billion times.
- A writer pays for breaking the strike in this "viral" video "Viral" video
- There's no evidence that Adblock Plus or similar products are about to go viral.
- Launched by Sean Parker (a Zuckerberg pal and partner in the Founders Fund, which owns a stake in Facebook) and Joe Green, the application allows Facebook users to tap into the latent viral power of their social networks for good causes.
- Add in the viral nature of the contest and it proved to be a winning formula.
- Loiacono, a Warcraft fan, was looking to create some viral buzz for his start-up, a social networking site called Voig aimed at bringing video game fans together.
- The four artists left standing will battle it out for the (not contagious) "Viral Woodie."

Cluster: 1

- The investigational vaccine used in both ... studies cannot cause HIV infection because it contains only synthetically produced snippets of viral material.
- A viral heart infection called cardiomyopathy is to blame, People magazine reports.
- Two other wards at the hospital have closed to new admissions because three patients have suspected viral gastroenteritis.
- Using the company's proprietary virus-like particle (VLP) and Novasome (R) adjuvant technologies, Novavax is developing vaccines to protect against H5N1 pandemic influenza, seasonal flu and other viral diseases.
- treatment options (which could include surgery, radiation, chemotherapy, transplants, clinical trials, biological, hormonal, viral and complementary therapies and alternative therapies),
- Idenix Pharmaceuticals, Inc., headquartered in Cambridge, Massachusetts, is a biopharmaceutical company engaged in the discovery, development and commercialization of drugs for the treatment of human viral and other infectious diseases.
- Meningococcal disease often begins with symptoms that can be mistaken for common viral illnesses, such as the flu.
- The animal which tested positive had one of the highest viral counts ever observed in a red squirrel, said Dr Shuttleworth, and it was likely that other wild red squirrels would have been exposed to the virus.

Clustering Comparison

Algorithm	Silhouette Score ↑	Noise Ratio ↓	#Clusters
K-Means	0.62 (High)	4.5%	2
DBSCAN	0.21 (Low)	35% (High noise)	1–3
HDBSCAN	0.38 (Medium)	10%	5–8

Analysis

 Semantic drift reflects a shift in dominant usage, not full replacement

Ex: "viral" shifted from medical → internet context

 Different evolution patterns:

Semantic shift (viral)

Semantic expansion (mouse)

 Social media accelerates language change

Reddit shows faster emergence of new meanings

 Embedding + clustering approach captures semantic differences effectively

 Drift score measures contextual change over time

Limitations

Polysemy challenge:

- Words have multiple meanings (e.g., "mouse", "viral")
- Hard to separate true drift vs co-existing meanings

Data limitations:

- News and Reddit may not fully represent all language usage
- Temporal coverage may be uneven

Context imbalance:

- More Reddit data than historical data can bias results

Future Work

→ Use time-aware embeddings for better temporal modeling

→ Improve clustering for clearer meaning separation

→ Extend to more words and domains

→ Apply to real-time drift detection in NLP systems



THANKS!

